

GEV fit complete.

Datasets / San Diego – PRCP / GEV: San Diego – PRCP GEV

San Diego - PRCP GEV

Success · created 2026-05-17 01:17 · finished 2026-05-17 01:17

Run again

Delete

Years 86 1940–2025	Location (μ) 1.291 in	Scale (σ) 0.437 in	Shape (ξ) -0.1051 Weibull (bounded)	Log-likelihood -59.84 converged
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REFERENCE EXCEEDANCE

P(annual max $\geq$ **1.35 in** in any given year) = **0.5821** · expected return period $\approx$ **1.7 years** (reference: 2025 observed maximum)

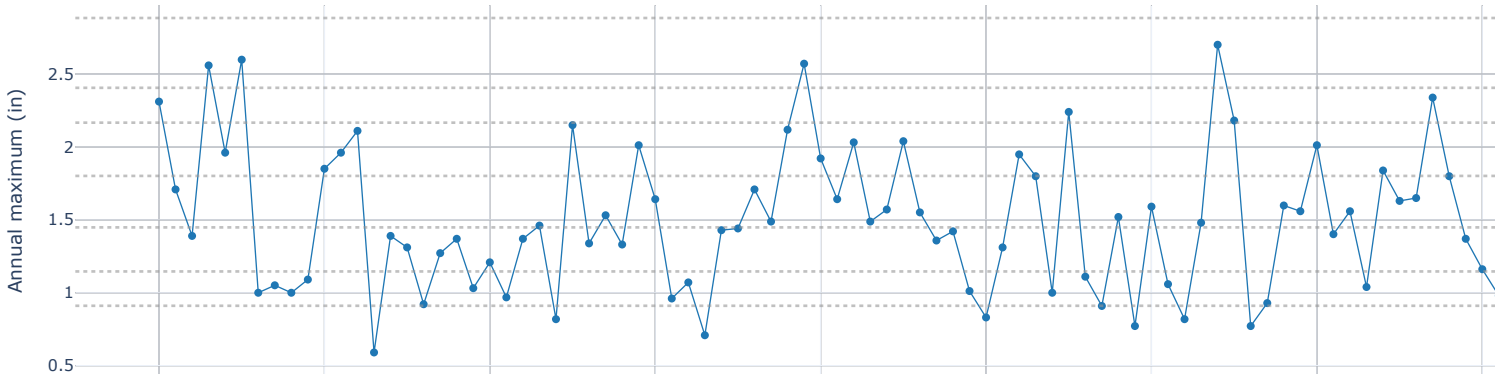
95% CONFIDENCE INTERVALS

Method	Iterations	Successful refits	Location (μ)	Scale (σ)	Shape (ξ)
Parametric	500	500	[1.197, 1.393] SD 0.050	[0.355, 0.511] SD 0.040	[-0.2651, 0.0473] SD 0.0782
Nonparametric	500	500	[1.179, 1.427] SD 0.061	[0.359, 0.508] SD 0.038	[-0.2460, 0.0795] SD 0.0826

RETURN LEVELS

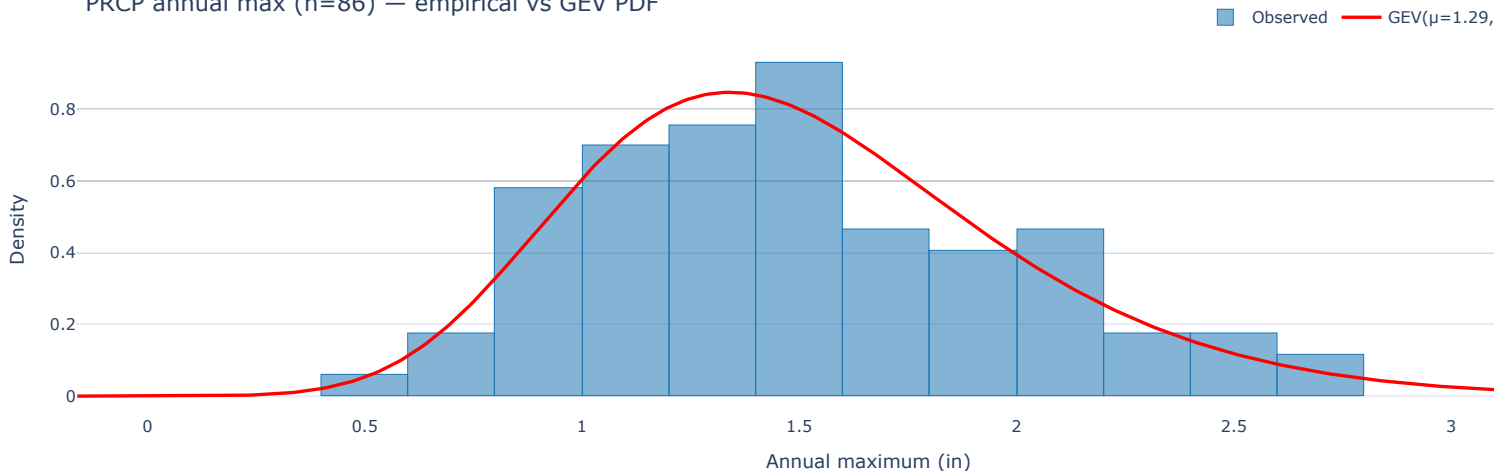
Return period (years)	Return level (in)
2	1.448
5	1.897
10	2.166
25	2.478
50	2.689
100	2.884
200	3.065
500	3.284
1000	3.436

PRCP annual max (n=86) with fitted percentiles



Year

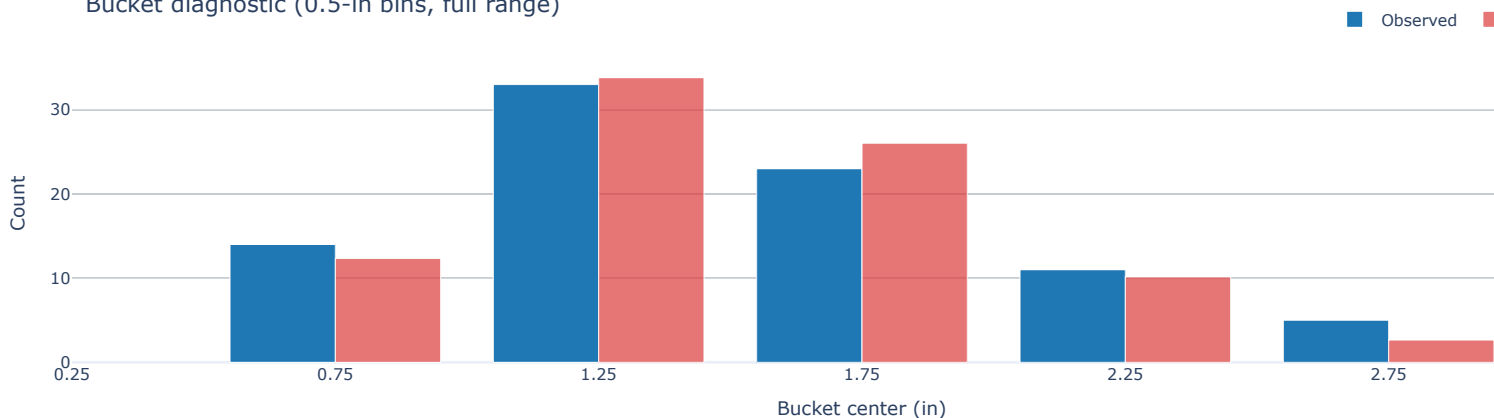
PRCP annual max (n=86) — empirical vs GEV PDF



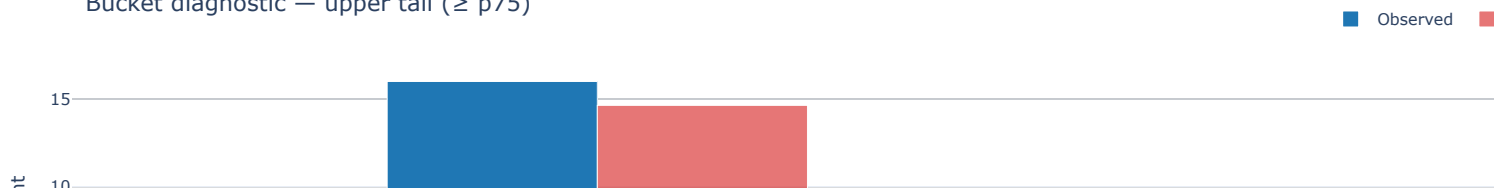
Return-level plot with 95% parametric CI

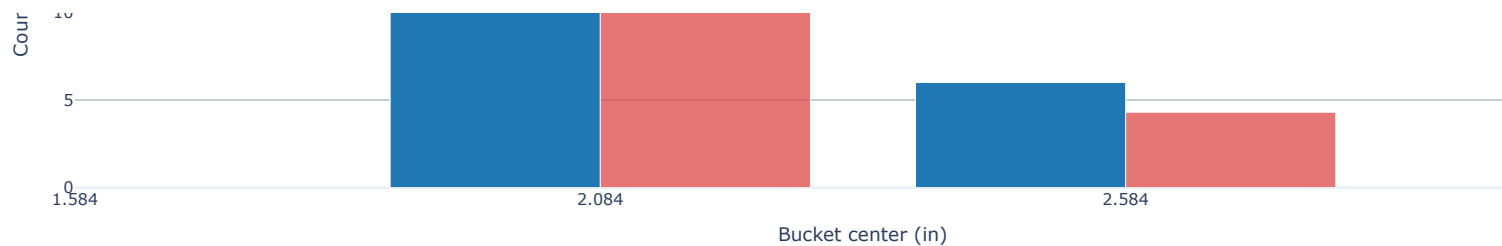


Bucket diagnostic (0.5-in bins, full range)

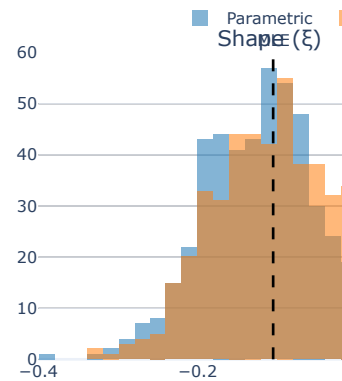
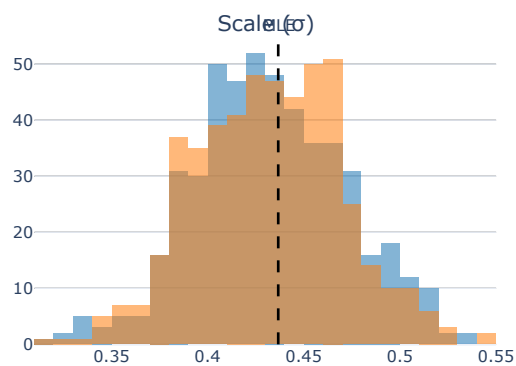
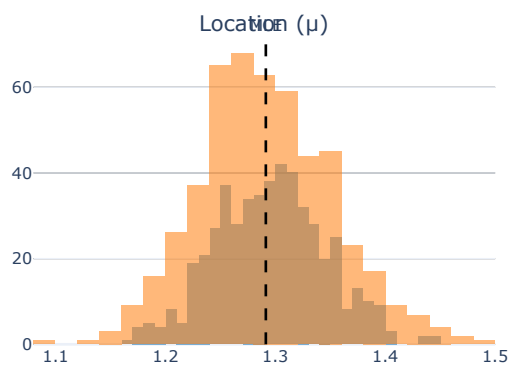


Bucket diagnostic — upper tail (≥ p75)





Bootstrap parameter distributions



► Parameters used

WeatherExtremes · Non-stationary occurrence-rate analysis of extreme events.